Database Management Systems - Week 5

DML and Join SQL Queries

Smart Hostel Management System - A Case Study

SRN: PES1UG24AM387

Contents: (1) schema creation screenshots, (2) inserted data screenshots, (3) Part A questions 1-10 and Part B questions 1-12, each with its SQL query and output. SQL file: PES1UG24AM387_Lab04_DML_Join.sql.

Task 1 : Database Preparation - Schema creation

Set the prompt

```
MySQL command line client - PES1UG24AM387

MariaDB [(none)]> prompt (PES1UG24AM387@\u) [\d]>\_
PROMPT set to '(PES1UG24AM387@\u) [\d]>\_'
(PES1UG24AM387@root) [(none)]>
```

Create the database

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [(none)]> DROP DATABASE IF EXISTS smart_hostel;
Query OK, 0 rows affected, 1 warning (0.000 sec)

(PES1UG24AM387@root) [(none)]> CREATE DATABASE smart_hostel;
Query OK, 1 row affected (0.001 sec)

(PES1UG24AM387@root) [(none)]> USE smart_hostel;
Database changed
(PES1UG24AM387@root) [smart_hostel]>
```

Create HOSTEL, STAFF, ROOM, BED

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> CREATE TABLE hostel (
    ->     Hostel_ID    INT          NOT NULL,
    ->     Hostel_Name  VARCHAR(50)  NOT NULL,
    ->     Location     VARCHAR(100) NOT NULL,
    ->     Capacity     INT          NOT NULL DEFAULT 0,
    ->     CONSTRAINT pk_hostel PRIMARY KEY (Hostel_ID),
    ->     CONSTRAINT chk_hostel_capacity CHECK (Capacity >= 0)
    -> );
Query OK, 0 rows affected, 1 warning (0.006 sec)

(PES1UG24AM387@root) [smart_hostel]> CREATE TABLE staff (
    ->     Staff_ID       INT           NOT NULL,
    ->     Staff_Name     VARCHAR(50)   NOT NULL,
    ->     Salary         DECIMAL(10,2) NOT NULL,
    ->     Designation    VARCHAR(30)   NOT NULL DEFAULT 'Support Staff',
    ->     Hostel_ID      INT           NOT NULL,
    ->     Supervisor_ID  INT           NULL,
    ->     CONSTRAINT pk_staff PRIMARY KEY (Staff_ID),
    ->     CONSTRAINT chk_staff_salary CHECK (Salary > 0),
    ->     CONSTRAINT fk_staff_hostel FOREIGN KEY (Hostel_ID) REFERENCES hostel  (Hostel_ID),
    ->     CONSTRAINT fk_staff_supervisor FOREIGN KEY (Supervisor_ID)
    ->         REFERENCES staff (Staff_ID) ON DELETE SET NULL
    -> );
Query OK, 0 rows affected, 1 warning (0.007 sec)

(PES1UG24AM387@root) [smart_hostel]> CREATE TABLE room (
    ->     Hostel_ID    INT NOT NULL,
    ->     Room_Number  INT NOT NULL,
    ->     Floor        INT NOT NULL DEFAULT 0,
    ->     CONSTRAINT pk_room PRIMARY KEY (Hostel_ID, Room_Number),
    ->     CONSTRAINT fk_room_hostel FOREIGN KEY (Hostel_ID)
    ->         REFERENCES hostel (Hostel_ID) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.008 sec)
```

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> CREATE TABLE bed (
    ->     Hostel_ID    INT         NOT NULL,
    ->     Room_Number  INT         NOT NULL,
    ->     Bed_Number   INT         NOT NULL,
    ->     Status       VARCHAR(10) NOT NULL DEFAULT 'Vacant',
    ->     CONSTRAINT pk_bed PRIMARY KEY (Hostel_ID, Room_Number, Bed_Number),
    ->     CONSTRAINT chk_bed_status CHECK (Status IN ('Occupied', 'Vacant')),
    ->     CONSTRAINT fk_bed_room FOREIGN KEY (Hostel_ID, Room_Number)
    ->         REFERENCES room (Hostel_ID, Room_Number) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.006 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Create STUDENT, STUDENT_PHONE, FACILITY, USES

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> CREATE TABLE student (
    ->     Student_ID   INT         NOT NULL,
    ->     First_Name   VARCHAR(30) NOT NULL,
    ->     Middle_Name  VARCHAR(30) NULL,
    ->     Last_Name    VARCHAR(30) NOT NULL,
    ->     Gender       CHAR(1)     NOT NULL,
    ->     DOB          DATE        NOT NULL,
    ->     Email        VARCHAR(60) NOT NULL,
    ->     CONSTRAINT pk_student PRIMARY KEY (Student_ID),
    ->     CONSTRAINT uq_student_email UNIQUE (Email),
    ->     CONSTRAINT chk_student_gender CHECK (Gender IN ('M', 'F', 'O'))
    -> );
Query OK, 0 rows affected, 1 warning (0.005 sec)

(PES1UG24AM387@root) [smart_hostel]> CREATE TABLE student_phone (
    ->     Student_ID    INT         NOT NULL,
    ->     Phone_Number  VARCHAR(15) NOT NULL,
    ->     CONSTRAINT pk_student_phone PRIMARY KEY (Student_ID, Phone_Number),
    ->     CONSTRAINT fk_phone_student FOREIGN KEY (Student_ID)
    ->         REFERENCES student (Student_ID) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.011 sec)

(PES1UG24AM387@root) [smart_hostel]> CREATE TABLE facility (
    ->     Facility_ID      INT         NOT NULL,
    ->     Facility_Name    VARCHAR(40) NOT NULL,
    ->     Operating_Hours  VARCHAR(30) DEFAULT '24x7',
    ->     Hostel_ID        INT         NOT NULL,
    ->     CONSTRAINT pk_facility PRIMARY KEY (Facility_ID),
    ->     CONSTRAINT fk_facility_hostel FOREIGN KEY (Hostel_ID)
    ->         REFERENCES hostel (Hostel_ID) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.006 sec)

(PES1UG24AM387@root) [smart_hostel]> CREATE TABLE uses (
    ->     Student_ID   INT  NOT NULL,
    ->     Facility_ID  INT  NOT NULL,
    ->     Usage_Date   DATE NOT NULL,
    ->     CONSTRAINT pk_uses PRIMARY KEY (Student_ID, Facility_ID, Usage_Date),
    ->     CONSTRAINT fk_uses_student FOREIGN KEY (Student_ID)
    ->         REFERENCES student (Student_ID) ON DELETE CASCADE,
    ->     CONSTRAINT fk_uses_facility FOREIGN KEY (Facility_ID)
    ->         REFERENCES facility (Facility_ID) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.005 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Create ALLOCATION, PAYMENT, RECEIPT

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> CREATE TABLE allocation (
    ->     Allocation_ID    INT  NOT NULL,
    ->     Student_ID       INT  NOT NULL,
    ->     Hostel_ID        INT  NOT NULL,
    ->     Room_Number      INT  NOT NULL,
    ->     Allocation_Date  DATE NOT NULL,
    ->     Check_in_Date    DATE NOT NULL,
    ->     Check_out_Date   DATE NULL,
    ->     CONSTRAINT pk_allocation PRIMARY KEY (Allocation_ID),
    ->     CONSTRAINT fk_alloc_student FOREIGN KEY (Student_ID)
    ->         REFERENCES student (Student_ID) ON DELETE CASCADE,
    ->     CONSTRAINT fk_alloc_room FOREIGN KEY (Hostel_ID, Room_Number)
    ->         REFERENCES room (Hostel_ID, Room_Number) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.007 sec)

(PES1UG24AM387@root) [smart_hostel]> CREATE TABLE payment (
    ->     Payment_ID    INT           NOT NULL,
    ->     Student_ID    INT           NOT NULL,
    ->     Payment_Date  DATE          NOT NULL,
    ->     Amount        DECIMAL(10,2) NOT NULL,
    ->     Payment_Type  VARCHAR(20)   NOT NULL,
    ->     CONSTRAINT pk_payment PRIMARY KEY (Payment_ID),
    ->     CONSTRAINT chk_payment_amount CHECK (Amount > 0),
    ->     CONSTRAINT fk_payment_student FOREIGN KEY (Student_ID) REFERENCES stu dent (Student_ID)
    -> );
Query OK, 0 rows affected, 1 warning (0.006 sec)

(PES1UG24AM387@root) [smart_hostel]> CREATE TABLE receipt (
    ->     Receipt_ID    INT  NOT NULL,
    ->     Payment_ID    INT  NOT NULL,
    ->     Receipt_Date  DATE NOT NULL,
    ->     CONSTRAINT pk_receipt PRIMARY KEY (Receipt_ID),
    ->     CONSTRAINT uq_receipt_payment UNIQUE (Payment_ID),
    ->     CONSTRAINT fk_receipt_payment FOREIGN KEY (Payment_ID) REFERENCES pay ment (Payment_ID)
    -> );
Query OK, 0 rows affected, 1 warning (0.007 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Verify the schema (SHOW TABLES, DESC)

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SHOW TABLES;
+------------------------+
| Tables_in_smart_hostel |
+------------------------+
| allocation             |
| bed                    |
| facility               |
| hostel                 |
| payment                |
| receipt                |
| room                   |
| staff                  |
| student                |
| student_phone          |
| uses                   |
+------------------------+
11 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> DESC hostel;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| Hostel_ID   | int(11)      | NO   | PRI | NULL    |       |
| Hostel_Name | varchar(50)  | NO   |     | NULL    |       |
| Location    | varchar(100) | NO   |     | NULL    |       |
| Capacity    | int(11)      | NO   |     | 0       |       |
+-------------+--------------+------+-----+---------+-------+
4 rows in set (0.002 sec)

(PES1UG24AM387@root) [smart_hostel]> DESC staff;
+---------------+---------------+------+-----+---------------+-------+
| Field         | Type          | Null | Key | Default       | Extra |
+---------------+---------------+------+-----+---------------+-------+
| Staff_ID      | int(11)       | NO   | PRI | NULL          |       |
| Staff_Name    | varchar(50)   | NO   |     | NULL          |       |
| Salary        | decimal(10,2) | NO   |     | NULL          |       |
| Designation   | varchar(30)   | NO   |     | Support Staff |       |
| Hostel_ID     | int(11)       | NO   | MUL | NULL          |       |
| Supervisor_ID | int(11)       | YES  | MUL | NULL          |       |
+---------------+---------------+------+-----+---------------+-------+
6 rows in set (0.001 sec)
```

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> DESC room;
+-------------+---------+------+-----+---------+-------+
| Field       | Type    | Null | Key | Default | Extra |
+-------------+---------+------+-----+---------+-------+
| Hostel_ID   | int(11) | NO   | PRI | NULL    |       |
| Room_Number | int(11) | NO   | PRI | NULL    |       |
| Floor       | int(11) | NO   |     | 0       |       |
+-------------+---------+------+-----+---------+-------+
3 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> DESC bed;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| Hostel_ID   | int(11)     | NO   | PRI | NULL    |       |
| Room_Number | int(11)     | NO   | PRI | NULL    |       |
| Bed_Number  | int(11)     | NO   | PRI | NULL    |       |
| Status      | varchar(10) | NO   |     | Vacant  |       |
+-------------+-------------+------+-----+---------+-------+
4 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> DESC student;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| Student_ID  | int(11)     | NO   | PRI | NULL    |       |
| First_Name  | varchar(30) | NO   |     | NULL    |       |
| Middle_Name | varchar(30) | YES  |     | NULL    |       |
| Last_Name   | varchar(30) | NO   |     | NULL    |       |
| Gender      | char(1)     | NO   |     | NULL    |       |
| DOB         | date        | NO   |     | NULL    |       |
| Email       | varchar(60) | NO   | UNI | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
7 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> DESC student_phone;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| Student_ID   | int(11)     | NO   | PRI | NULL    |       |
| Phone_Number | varchar(15) | NO   | PRI | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
2 rows in set (0.001 sec)
```

MySQL command line client - PES1UG24AM387

```
(PES1UG24AM387@root) [smart_hostel]> DESC facility;
+-----------------+-------------+------+-----+---------+-------+
| Field           | Type        | Null | Key | Default | Extra |
+-----------------+-------------+------+-----+---------+-------+
| Facility_ID     | int(11)     | NO   | PRI | NULL    |       |
| Facility_Name   | varchar(40) | NO   |     | NULL    |       |
| Operating_Hours | varchar(30) | YES  |     | 24x7    |       |
| Hostel_ID       | int(11)     | NO   | MUL | NULL    |       |
+-----------------+-------------+------+-----+---------+-------+
4 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> DESC uses;
+-------------+---------+------+-----+---------+-------+
| Field       | Type    | Null | Key | Default | Extra |
+-------------+---------+------+-----+---------+-------+
| Student_ID  | int(11) | NO   | PRI | NULL    |       |
| Facility_ID | int(11) | NO   | PRI | NULL    |       |
| Usage_Date  | date    | NO   | PRI | NULL    |       |
+-------------+---------+------+-----+---------+-------+
3 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> DESC allocation;
+-----------------+---------+------+-----+---------+-------+
| Field           | Type    | Null | Key | Default | Extra |
+-----------------+---------+------+-----+---------+-------+
| Allocation_ID   | int(11) | NO   | PRI | NULL    |       |
| Student_ID      | int(11) | NO   | MUL | NULL    |       |
| Hostel_ID       | int(11) | NO   | MUL | NULL    |       |
| Room_Number     | int(11) | NO   |     | NULL    |       |
| Allocation_Date | date    | NO   |     | NULL    |       |
| Check_in_Date   | date    | NO   |     | NULL    |       |
| Check_out_Date  | date    | YES  |     | NULL    |       |
+-----------------+---------+------+-----+---------+-------+
7 rows in set (0.001 sec)
```

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> DESC payment;
+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| Payment_ID   | int(11)       | NO   | PRI | NULL    |       |
| Student_ID   | int(11)       | NO   | MUL | NULL    |       |
| Payment_Date | date          | NO   |     | NULL    |       |
| Amount       | decimal(10,2) | NO   |     | NULL    |       |
| Payment_Type | varchar(20)   | NO   |     | NULL    |       |
+--------------+---------------+------+-----+---------+-------+
5 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> DESC receipt;
+--------------+---------+------+-----+---------+-------+
| Field        | Type    | Null | Key | Default | Extra |
+--------------+---------+------+-----+---------+-------+
| Receipt_ID   | int(11) | NO   | PRI | NULL    |       |
| Payment_ID   | int(11) | NO   | UNI | NULL    |       |
| Receipt_Date | date    | NO   |     | NULL    |       |
+--------------+---------+------+-----+---------+-------+
3 rows in set (0.002 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Task 1 : Inserted data

Populate the tables with the sample data

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO hostel (Hostel_ID, Hostel_Name,  Location, Capacity) VALUES
    ->     (1, 'Sunrise Hostel', 'North Campus', 120),
    ->     (2, 'Blue Ridge Hostel', 'South Campus', 90),
    ->     (3, 'Lake View Hostel', 'East Campus', 60);
Query OK, 3 rows affected (0.002 sec)
Records: 3  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO staff (Staff_ID, Staff_Name, De signation, Salary, Hostel_ID, Supervisor_ID) VALUES
    ->     (1, 'Ramesh Iyer', 'Warden', 45000, 1, NULL),
    ->     (2, 'Sunil Kumar', 'Security', 22000, 1, 1),
    ->     (3, 'Geeta Rao', 'Cleaner', 18000, 1, 1),
    ->     (4, 'Anita Sharma', 'Warden', 46000, 2, NULL),
    ->     (5, 'Mohan Das', 'Security', 21000, 2, 4),
    ->     (6, 'Priya Menon', 'Maintenance', 25000, 2, 4),
    ->     (7, 'Kiran Bose', 'Warden', 44000, 3, NULL),
    ->     (8, 'Farhan Sheikh', 'Cleaner', 17500, 3, 7);
Query OK, 8 rows affected (0.002 sec)
Records: 8  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO room (Hostel_ID, Room_Number, F loor) VALUES
    ->     (1, 101, 1), (1, 102, 1), (1, 201, 2),
    ->     (2, 101, 1), (2, 102, 1),
    ->     (3, 101, 1);
Query OK, 6 rows affected (0.003 sec)
Records: 6  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO bed (Hostel_ID, Room_Number, Be d_Number, Status) VALUES
    ->     (1, 101, 1, 'Occupied'), (1, 101, 2, 'Occupied'),
    ->     (1, 102, 1, 'Occupied'), (1, 102, 2, 'Vacant'),
    ->     (1, 201, 1, 'Occupied'), (1, 201, 2, 'Vacant'),
    ->     (2, 101, 1, 'Occupied'), (2, 101, 2, 'Vacant'),
    ->     (2, 102, 1, 'Occupied'), (2, 102, 2, 'Occupied'),
    ->     (3, 101, 1, 'Occupied'), (3, 101, 2, 'Vacant');
Query OK, 12 rows affected (0.003 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO student (Student_ID, First_Name , Middle_Name, Last_Name, Gender, DOB, Email) VALUES
    ->     (1, 'Aditya', NULL, 'Verma', 'M', '2005-03-14', 'aditya.verma@pes.edu '),
    ->     (2, 'Neha', 'R', 'Kulkarni', 'F', '2005-07-22', 'neha.kulkarni@pes.ed u'),
    ->     (3, 'Rohan', NULL, 'Shetty', 'M', '2004-11-05', 'rohan.shetty@pes.edu '),
    ->     (4, 'Sanya', 'K', 'Malhotra', 'F', '2005-01-30', 'sanya.malhotra@pes. edu'),
    ->     (5, 'Vikram', NULL, 'Nair', 'M', '2004-09-18', 'vikram.nair@pes.edu') ,
    ->     (6, 'Ishita', 'P', 'Bansal', 'F', '2005-05-09', 'ishita.bansal@pes.ed u'),
    ->     (7, 'Arjun', NULL, 'Reddy', 'M', '2005-02-27', 'arjun.reddy@pes.edu') ,
    ->     (8, 'Meera', 'S', 'Pillai', 'F', '2004-12-12', 'meera.pillai@pes.edu' );
Query OK, 8 rows affected (0.002 sec)
Records: 8  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO student_phone (Student_ID, Phon e_Number) VALUES
    ->     (1, '9876500001'), (1, '9876500011'), (2, '9876500002'), (3, '9876500 003'),
    ->     (4, '9876500004'), (4, '9876500014'), (5, '9876500005'), (6, '9876500 006'),
    ->     (7, '9876500007'), (8, '9876500008');
Query OK, 10 rows affected (0.004 sec)
Records: 10  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO facility (Facility_ID, Facility _Name, Operating_Hours, Hostel_ID) VALUES
    ->     (1, 'WiFi Lounge', '24 Hours', 1),
    ->     (2, 'Laundry Room', '06:00-20:00', 1),
    ->     (3, 'Gym', '05:00-22:00', 2),
    ->     (4, 'Study Room', '24 Hours', 2),
    ->     (5, 'Common Room', '08:00-23:00', 3);
Query OK, 5 rows affected (0.003 sec)
Records: 5  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO uses (Student_ID, Facility_ID,  Usage_Date) VALUES
    ->     (1, 1, '2026-01-20'), (2, 1, '2026-01-20'), (3, 1, '2026-01-22'),
    ->     (1, 2, '2026-01-21'), (4, 3, '2026-01-20'), (5, 3, '2026-01-21'),
    ->     (4, 4, '2026-01-25'), (6, 4, '2026-01-22'), (7, 5, '2026-01-23');
Query OK, 9 rows affected (0.002 sec)
Records: 9  Duplicates: 0  Warnings: 0
```

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO allocation (Allocation_ID, Stud ent_ID, Hostel_ID, Room_Number,
    ->                          Allocation_Date, Check_in_Date, Check_out_Date) V ALUES
    ->     (1, 1, 1, 101, '2026-01-05', '2026-01-06', NULL),
    ->     (2, 2, 1, 101, '2026-01-05', '2026-01-06', NULL),
    ->     (3, 3, 1, 102, '2026-01-06', '2026-01-07', NULL),
    ->     (4, 4, 2, 101, '2026-01-05', '2026-01-06', NULL),
    ->     (5, 5, 2, 102, '2026-01-06', '2026-01-07', NULL),
    ->     (6, 6, 2, 102, '2026-01-06', '2026-01-07', NULL),
    ->     (7, 7, 3, 101, '2026-01-07', '2026-01-08', NULL),
    ->     (8, 3, 1, 201, '2025-06-01', '2025-06-02', '2025-12-20');
Query OK, 8 rows affected (0.002 sec)
Records: 8  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO payment (Payment_ID, Student_ID , Payment_Date, Amount, Payment_Type) VALUES
    ->     (1, 1, '2026-01-10', 45000, 'Online'),
    ->     (2, 2, '2026-01-11', 45000, 'Cash'),
    ->     (3, 3, '2026-01-12', 42000, 'Card'),
    ->     (4, 4, '2026-01-10', 48000, 'Online'),
    ->     (5, 5, '2026-01-13', 48000, 'Cheque'),
    ->     (6, 6, '2026-01-14', 48000, 'Online'),
    ->     (7, 7, '2026-01-15', 40000, 'Cash'),
    ->     (8, 1, '2026-06-01', 45000, 'Online');
Query OK, 8 rows affected (0.002 sec)
Records: 8  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO receipt (Receipt_ID, Payment_ID , Receipt_Date) VALUES
    ->     (1, 1, '2026-01-10'), (2, 2, '2026-01-11'), (3, 3, '2026-01-12'),
    ->     (4, 4, '2026-01-10'), (5, 5, '2026-01-13'), (6, 6, '2026-01-14'),
    ->     (7, 7, '2026-01-15');
Query OK, 7 rows affected (0.003 sec)
Records: 7  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]>
```

Verify the data: HOSTEL, STAFF, ROOM, BED

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM hostel;
+-----------+-------------------+--------------+----------+
| Hostel_ID | Hostel_Name       | Location     | Capacity |
+-----------+-------------------+--------------+----------+
|         1 | Sunrise Hostel    | North Campus |      120 |
|         2 | Blue Ridge Hostel | South Campus |       90 |
|         3 | Lake View Hostel  | East Campus  |       60 |
+-----------+-------------------+--------------+----------+
3 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM staff;
+----------+---------------+----------+-------------+-----------+---------------+
| Staff_ID | Staff_Name    | Salary   | Designation | Hostel_ID | Supervisor_ID |
+----------+---------------+----------+-------------+-----------+---------------+
|        1 | Ramesh Iyer   | 45000.00 | Warden      |         1 |          NULL |
|        2 | Sunil Kumar   | 22000.00 | Security    |         1 |             1 |
|        3 | Geeta Rao     | 18000.00 | Cleaner     |         1 |             1 |
|        4 | Anita Sharma  | 46000.00 | Warden      |         2 |          NULL |
|        5 | Mohan Das     | 21000.00 | Security    |         2 |             4 |
|        6 | Priya Menon   | 25000.00 | Maintenance |         2 |             4 |
|        7 | Kiran Bose    | 44000.00 | Warden      |         3 |          NULL |
|        8 | Farhan Sheikh | 17500.00 | Cleaner     |         3 |             7 |
+----------+---------------+----------+-------------+-----------+---------------+
8 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM room;
+-----------+-------------+-------+
| Hostel_ID | Room_Number | Floor |
+-----------+-------------+-------+
|         1 |         101 |     1 |
|         1 |         102 |     1 |
|         1 |         201 |     2 |
|         2 |         101 |     1 |
|         2 |         102 |     1 |
|         3 |         101 |     1 |
+-----------+-------------+-------+
6 rows in set (0.001 sec)
```

MySQL command line client - PES1UG24AM387

```
(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM bed;
+-----------+-------------+------------+----------+
| Hostel_ID | Room_Number | Bed_Number | Status   |
+-----------+-------------+------------+----------+
|         1 |         101 |          1 | Occupied |
|         1 |         101 |          2 | Occupied |
|         1 |         102 |          1 | Occupied |
|         1 |         102 |          2 | Vacant   |
|         1 |         201 |          1 | Occupied |
|         1 |         201 |          2 | Vacant   |
|         2 |         101 |          1 | Occupied |
|         2 |         101 |          2 | Vacant   |
|         2 |         102 |          1 | Occupied |
|         2 |         102 |          2 | Occupied |
|         3 |         101 |          1 | Occupied |
|         3 |         101 |          2 | Vacant   |
+-----------+-------------+------------+----------+
12 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Verify the data: STUDENT, STUDENT_PHONE, FACILITY, USES

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM student;
+------------+------------+-------------+-----------+--------+------------+------------------------+
| Student_ID | First_Name | Middle_Name | Last_Name | Gender | DOB        | Email                  |
+------------+------------+-------------+-----------+--------+------------+------------------------+
|          1 | Aditya     | NULL        | Verma     | M      | 2005-03-14 | aditya.verma@pes.edu   |
|          2 | Neha       | R           | Kulkarni  | F      | 2005-07-22 | neha.kulkarni@pes.edu  |
|          3 | Rohan      | NULL        | Shetty    | M      | 2004-11-05 | rohan.shetty@pes.edu   |
|          4 | Sanya      | K           | Malhotra  | F      | 2005-01-30 | sanya.malhotra@pes.edu |
|          5 | Vikram     | NULL        | Nair      | M      | 2004-09-18 | vikram.nair@pes.edu    |
|          6 | Ishita     | P           | Bansal    | F      | 2005-05-09 | ishita.bansal@pes.edu  |
|          7 | Arjun      | NULL        | Reddy     | M      | 2005-02-27 | arjun.reddy@pes.edu    |
|          8 | Meera      | S           | Pillai    | F      | 2004-12-12 | meera.pillai@pes.edu   |
+------------+------------+-------------+-----------+--------+------------+------------------------+
8 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM student_phone;
+------------+--------------+
| Student_ID | Phone_Number |
+------------+--------------+
|          1 | 9876500001   |
|          1 | 9876500011   |
|          2 | 9876500002   |
|          3 | 9876500003   |
|          4 | 9876500004   |
|          4 | 9876500014   |
|          5 | 9876500005   |
|          6 | 9876500006   |
|          7 | 9876500007   |
|          8 | 9876500008   |
+------------+--------------+
10 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM facility;
+-------------+---------------+-----------------+-----------+
| Facility_ID | Facility_Name | Operating_Hours | Hostel_ID |
+-------------+---------------+-----------------+-----------+
|           1 | WiFi Lounge   | 24 Hours        |         1 |
|           2 | Laundry Room  | 06:00-20:00     |         1 |
|           3 | Gym           | 05:00-22:00     |         2 |
|           4 | Study Room    | 24 Hours        |         2 |
|           5 | Common Room   | 08:00-23:00     |         3 |
+-------------+---------------+-----------------+-----------+
5 rows in set (0.001 sec)
```

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM uses;
+------------+-------------+------------+
| Student_ID | Facility_ID | Usage_Date |
+------------+-------------+------------+
|          1 |           1 | 2026-01-20 |
|          1 |           2 | 2026-01-21 |
|          2 |           1 | 2026-01-20 |
|          3 |           1 | 2026-01-22 |
|          4 |           3 | 2026-01-20 |
|          4 |           4 | 2026-01-25 |
|          5 |           3 | 2026-01-21 |
|          6 |           4 | 2026-01-22 |
|          7 |           5 | 2026-01-23 |
+------------+-------------+------------+
9 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Verify the data: ALLOCATION, PAYMENT, RECEIPT

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM allocation;
+---------------+------------+-----------+-------------+-----------------+---------------+----------------+
| Allocation_ID | Student_ID | Hostel_ID | Room_Number | Allocation_Date | Check_in_Date | Check_out_Date |
+---------------+------------+-----------+-------------+-----------------+---------------+----------------+
|             1 |          1 |         1 |         101 | 2026-01-05      | 2026-01-06    | NULL           |
|             2 |          2 |         1 |         101 | 2026-01-05      | 2026-01-06    | NULL           |
|             3 |          3 |         1 |         102 | 2026-01-06      | 2026-01-07    | NULL           |
|             4 |          4 |         2 |         101 | 2026-01-05      | 2026-01-06    | NULL           |
|             5 |          5 |         2 |         102 | 2026-01-06      | 2026-01-07    | NULL           |
|             6 |          6 |         2 |         102 | 2026-01-06      | 2026-01-07    | NULL           |
|             7 |          7 |         3 |         101 | 2026-01-07      | 2026-01-08    | NULL           |
|             8 |          3 |         1 |         201 | 2025-06-01      | 2025-06-02    | 2025-12-20     |
+---------------+------------+-----------+-------------+-----------------+---------------+----------------+
8 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM payment;
+------------+------------+--------------+----------+--------------+
| Payment_ID | Student_ID | Payment_Date | Amount   | Payment_Type |
+------------+------------+--------------+----------+--------------+
|          1 |          1 | 2026-01-10   | 45000.00 | Online       |
|          2 |          2 | 2026-01-11   | 45000.00 | Cash         |
|          3 |          3 | 2026-01-12   | 42000.00 | Card         |
|          4 |          4 | 2026-01-10   | 48000.00 | Online       |
|          5 |          5 | 2026-01-13   | 48000.00 | Cheque       |
|          6 |          6 | 2026-01-14   | 48000.00 | Online       |
|          7 |          7 | 2026-01-15   | 40000.00 | Cash         |
|          8 |          1 | 2026-06-01   | 45000.00 | Online       |
+------------+------------+--------------+----------+--------------+
8 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM receipt;
+------------+------------+--------------+
| Receipt_ID | Payment_ID | Receipt_Date |
+------------+------------+--------------+
|          1 |          1 | 2026-01-10   |
|          2 |          2 | 2026-01-11   |
|          3 |          3 | 2026-01-12   |
|          4 |          4 | 2026-01-10   |
|          5 |          5 | 2026-01-13   |
|          6 |          6 | 2026-01-14   |
|          7 |          7 | 2026-01-15   |
+------------+------------+--------------+
7 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Task 2 : Part A - DML and Set Operations

Part A - Q1. Insert a new student, Kabir S. Oberoi (Male, DOB 2005-08-19, email kabir.oberoi@pes.edu), into Student.

SQL query:

```
INSERT INTO student (Student_ID, First_Name, Middle_Name, Last_Name, Gender, DOB, Email)
VALUES (9, 'Kabir', 'S', 'Oberoi', 'M', '2005-08-19', 'kabir.oberoi@pes.edu');

SELECT * FROM student WHERE Student_ID = 9;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO student (Student_ID, First_Name , Middle_Name, Last_Name, Gender, DOB, Email)
    -> VALUES (9, 'Kabir', 'S', 'Oberoi', 'M', '2005-08-19', 'kabir.oberoi@pes.e du');
Query OK, 1 row affected (0.002 sec)

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM student WHERE Student_ID = 9;
+------------+------------+-------------+-----------+--------+------------+----------------------+
| Student_ID | First_Name | Middle_Name | Last_Name | Gender | DOB        | Email                |
+------------+------------+-------------+-----------+--------+------------+----------------------+
|          9 | Kabir      | S           | Oberoi    | M      | 2005-08-19 | kabir.oberoi@pes.edu |
+------------+------------+-------------+-----------+--------+------------+----------------------+
1 row in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part A - Q2. In a single INSERT statement, add both of Kabir's phone numbers -- 9876500009 and 9876500019 -- to Student_Phone.

SQL query:

```
INSERT INTO student_phone (Student_ID, Phone_Number)
VALUES (9, '9876500009'), (9, '9876500019');

SELECT * FROM student_phone WHERE Student_ID = 9;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO student_phone (Student_ID, Phon e_Number)
    -> VALUES (9, '9876500009'), (9, '9876500019');
Query OK, 2 rows affected (0.001 sec)
Records: 2  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM student_phone WHERE Student_I D = 9;
+------------+--------------+
| Student_ID | Phone_Number |
+------------+--------------+
|          9 | 9876500009   |
|          9 | 9876500019   |
+------------+--------------+
2 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part A - Q3. In a single INSERT statement, record three new facility usage events: Kabir uses the WiFi Lounge (Facility_ID 1) on 2026-08-22, Kabir uses the Common Room (Facility_ID 5) on 2026-08-23, and Vikram Nair (Student_ID 5) uses the Gym (Facility_ID 3) on 2026-08-22.

SQL query:

```
INSERT INTO uses (Student_ID, Facility_ID, Usage_Date)
VALUES (9, 1, '2026-08-22'), (9, 5, '2026-08-23'), (5, 3, '2026-08-22');

SELECT * FROM uses WHERE Usage_Date >= '2026-08-01';
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO uses (Student_ID, Facility_ID,  Usage_Date)
    -> VALUES (9, 1, '2026-08-22'), (9, 5, '2026-08-23'), (5, 3, '2026-08-22');
Query OK, 3 rows affected (0.002 sec)
Records: 3  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM uses WHERE Usage_Date >= '202 6-08-01';
+------------+-------------+------------+
| Student_ID | Facility_ID | Usage_Date |
+------------+-------------+------------+
|          5 |           3 | 2026-08-22 |
|          9 |           1 | 2026-08-22 |
|          9 |           5 | 2026-08-23 |
+------------+-------------+------------+
3 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part A - Q4. The university approves a 10% salary hike for every Warden across all hostels. Update Staff accordingly.

SQL query:

```
UPDATE staff SET Salary = Salary * 1.10 WHERE Designation = 'Warden';

SELECT Staff_ID, Staff_Name, Designation, Salary FROM staff WHERE Designation = 'Warden';
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> UPDATE staff SET Salary = Salary * 1.10 WHE RE Designation = 'Warden';
Query OK, 3 rows affected (0.002 sec)
Rows matched: 3  Changed: 3  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> SELECT Staff_ID, Staff_Name, Designation, S alary FROM staff WHERE Designation = 'Warden';
+----------+--------------+-------------+----------+
| Staff_ID | Staff_Name   | Designation | Salary   |
+----------+--------------+-------------+----------+
|        1 | Ramesh Iyer  | Warden      | 49500.00 |
|        4 | Anita Sharma | Warden      | 50600.00 |
|        7 | Kiran Bose   | Warden      | 48400.00 |
+----------+--------------+-------------+----------+
3 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part A - Q5. Rohan Shetty (Student_ID 3) checks out of Room 102, Hostel 1 on 2026-08-20. Close his current allocation record and free up his bed (Bed_Number 1 in that room).

SQL query:

```
UPDATE allocation
SET Check_out_Date = '2026-08-20'
WHERE Student_ID = 3 AND Hostel_ID = 1 AND Room_Number = 102 AND Check_out_Date IS NULL;

UPDATE bed
SET Status = 'Vacant'
WHERE Hostel_ID = 1 AND Room_Number = 102 AND Bed_Number = 1;

SELECT * FROM allocation WHERE Student_ID = 3;

SELECT * FROM bed WHERE Hostel_ID = 1 AND Room_Number = 102;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> UPDATE allocation
    -> SET Check_out_Date = '2026-08-20'
    -> WHERE Student_ID = 3 AND Hostel_ID = 1 AND Room_Number = 102 AND Check_ou t_Date IS NULL;
Query OK, 1 row affected (0.004 sec)
Rows matched: 1  Changed: 1  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> UPDATE bed
    -> SET Status = 'Vacant'
    -> WHERE Hostel_ID = 1 AND Room_Number = 102 AND Bed_Number = 1;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM allocation WHERE Student_ID =  3;
+---------------+------------+-----------+-------------+-----------------+---------------+----------------+
| Allocation_ID | Student_ID | Hostel_ID | Room_Number | Allocation_Date | Check_in_Date | Check_out_Date |
+---------------+------------+-----------+-------------+-----------------+---------------+----------------+
|             3 |          3 |         1 |         102 | 2026-01-06      | 2026-01-07    | 2026-08-20     |
|             8 |          3 |         1 |         201 | 2025-06-01      | 2025-06-02    | 2025-12-20     |
+---------------+------------+-----------+-------------+-----------------+---------------+----------------+
2 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM bed WHERE Hostel_ID = 1 AND R oom_Number = 102;
+-----------+-------------+------------+--------+
| Hostel_ID | Room_Number | Bed_Number | Status |
+-----------+-------------+------------+--------+
|         1 |         102 |          1 | Vacant |
|         1 |         102 |          2 | Vacant |
+-----------+-------------+------------+--------+
2 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part A - Q6. Record a new payment of 45000 (Online) made by Kabir on 2026-08-22, and the receipt generated for it on the same date.

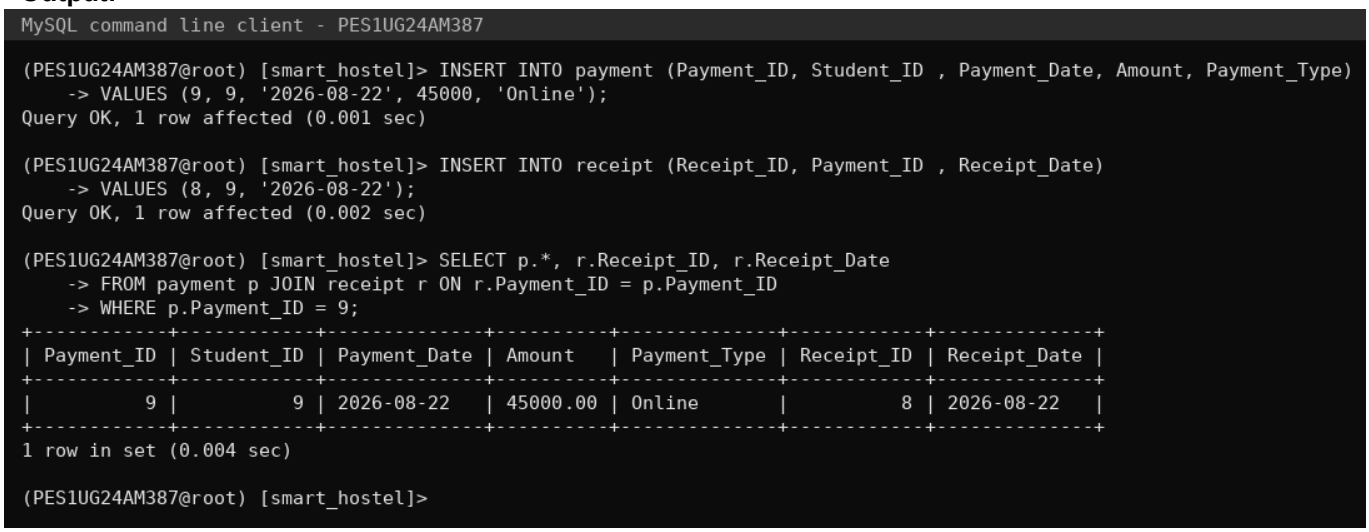
SQL query:

```
INSERT INTO payment (Payment_ID, Student_ID, Payment_Date, Amount, Payment_Type)
VALUES (9, 9, '2026-08-22', 45000, 'Online');

INSERT INTO receipt (Receipt_ID, Payment_ID, Receipt_Date)
VALUES (8, 9, '2026-08-22');

SELECT p.*, r.Receipt_ID, r.Receipt_Date
FROM payment p JOIN receipt r ON r.Payment_ID = p.Payment_ID
WHERE p.Payment_ID = 9;
```

Output:



```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO payment (Payment_ID, Student_ID , Payment_Date, Amount, Payment_Type)
    -> VALUES (9, 9, '2026-08-22', 45000, 'Online');
Query OK, 1 row affected (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]> INSERT INTO receipt (Receipt_ID, Payment_ID , Receipt_Date)
    -> VALUES (8, 9, '2026-08-22');
Query OK, 1 row affected (0.002 sec)

(PES1UG24AM387@root) [smart_hostel]> SELECT p.*, r.Receipt_ID, r.Receipt_Date
    -> FROM payment p JOIN receipt r ON r.Payment_ID = p.Payment_ID
    -> WHERE p.Payment_ID = 9;
+------------+------------+--------------+----------+--------------+------------+--------------+
| Payment_ID | Student_ID | Payment_Date | Amount   | Payment_Type | Receipt_ID | Receipt_Date |
+------------+------------+--------------+----------+--------------+------------+--------------+
|          9 |          9 | 2026-08-22   | 45000.00 | Online       |          8 | 2026-08-22   |
+------------+------------+--------------+----------+--------------+------------+--------------+
1 row in set (0.004 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part A - Q7. Delete every payment that does not yet have a matching receipt. Do this with a subquery -- do not hard-code any Payment_ID.

SQL query:

```
SELECT * FROM payment WHERE Payment_ID NOT IN (SELECT Payment_ID FROM receipt);

DELETE FROM payment
WHERE Payment_ID NOT IN (SELECT Payment_ID FROM receipt);

SELECT * FROM payment;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM payment WHERE Payment_ID NOT  IN (SELECT Payment_ID FROM receipt);
+------------+------------+--------------+----------+--------------+
| Payment_ID | Student_ID | Payment_Date | Amount   | Payment_Type |
+------------+------------+--------------+----------+--------------+
|          8 |          1 | 2026-06-01   | 45000.00 | Online       |
+------------+------------+--------------+----------+--------------+
1 row in set (0.002 sec)

(PES1UG24AM387@root) [smart_hostel]> DELETE FROM payment
    -> WHERE Payment_ID NOT IN (SELECT Payment_ID FROM receipt);
Query OK, 1 row affected (0.002 sec)

(PES1UG24AM387@root) [smart_hostel]> SELECT * FROM payment;
+------------+------------+--------------+----------+--------------+
| Payment_ID | Student_ID | Payment_Date | Amount   | Payment_Type |
+------------+------------+--------------+----------+--------------+
|          1 |          1 | 2026-01-10   | 45000.00 | Online       |
|          2 |          2 | 2026-01-11   | 45000.00 | Cash         |
|          3 |          3 | 2026-01-12   | 42000.00 | Card         |
|          4 |          4 | 2026-01-10   | 48000.00 | Online       |
|          5 |          5 | 2026-01-13   | 48000.00 | Cheque       |
|          6 |          6 | 2026-01-14   | 48000.00 | Online       |
|          7 |          7 | 2026-01-15   | 40000.00 | Cash         |
|          9 |          9 | 2026-08-22   | 45000.00 | Online       |
+------------+------------+--------------+----------+--------------+
8 rows in set (0.001 sec)

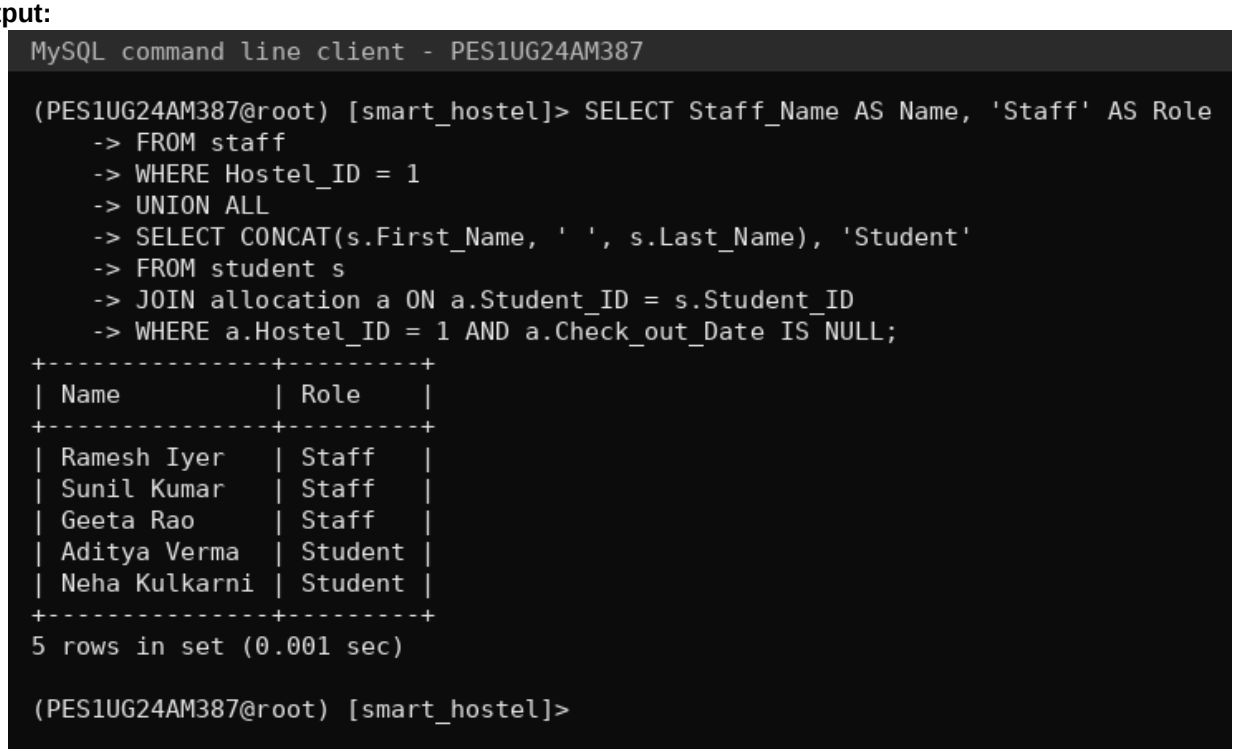
(PES1UG24AM387@root) [smart_hostel]>
```

Part A - Q8. List the names of all staff members working in Hostel 1 along with the first and last names of all students currently allocated to Hostel 1. Combine both result sets using UNION ALL. Display the name(s) and role (Staff or Student) for each record.

SQL query:

```
SELECT Staff_Name AS Name, 'Staff' AS Role
FROM staff
WHERE Hostel_ID = 1
UNION ALL
SELECT CONCAT(s.First_Name, ' ', s.Last_Name), 'Student'
FROM student s
JOIN allocation a ON a.Student_ID = s.Student_ID
WHERE a.Hostel_ID = 1 AND a.Check_out_Date IS NULL;
```

Output:



Part A - Q9. Using INTERSECT, find the Student_IDs of students who have both made at least one payment and used at least one facility.

SQL query:

```
SELECT Student_ID FROM payment
INTERSECT
SELECT Student_ID FROM uses
ORDER BY Student_ID;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT Student_ID FROM payment
    -> INTERSECT
    -> SELECT Student_ID FROM uses
    -> ORDER BY Student_ID;
+------------+
| Student_ID |
+------------+
|          1 |
|          2 |
|          3 |
|          4 |
|          5 |
|          6 |
|          7 |
|          9 |
+------------+
8 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part A - Q10. Using EXCEPT, find every student who has never made a payment.

SQL query:

```
SELECT Student_ID, First_Name, Last_Name FROM student
EXCEPT
SELECT s.Student_ID, s.First_Name, s.Last_Name
FROM student s JOIN payment p ON p.Student_ID = s.Student_ID;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT Student_ID, First_Name, Last_Name FR OM student
    -> EXCEPT
    -> SELECT s.Student_ID, s.First_Name, s.Last_Name
    -> FROM student s JOIN payment p ON p.Student_ID = s.Student_ID;
+------------+------------+-----------+
| Student_ID | First_Name | Last_Name |
+------------+------------+-----------+
|          8 | Meera      | Pillai    |
+------------+------------+-----------+
1 row in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Task 3 : Part B - Join Queries

Part B - Q1. List the Student_ID, first, middle and last name, hostel name and room number of every student who currently has an active allocation (Check_out_Date IS NULL).

SQL query:

```
SELECT s.Student_ID, s.First_Name, s.Middle_Name, s.Last_Name, h.Hostel_Name, a.Room_Number
FROM student s
INNER JOIN allocation a ON a.Student_ID = s.Student_ID
INNER JOIN hostel h ON h.Hostel_ID = a.Hostel_ID
WHERE a.Check_out_Date IS NULL
ORDER BY s.Student_ID;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT s.Student_ID, s.First_Name, s.Middle _Name, s.Last_Name, h.Hostel_Name, a.Room_Number
    -> FROM student s
    -> INNER JOIN allocation a ON a.Student_ID = s.Student_ID
    -> INNER JOIN hostel h ON h.Hostel_ID = a.Hostel_ID
    -> WHERE a.Check_out_Date IS NULL
    -> ORDER BY s.Student_ID;
+------------+------------+-------------+-----------+------------------+-------------+
| Student_ID | First_Name | Middle_Name | Last_Name | Hostel_Name      | Room_Number |
+------------+------------+-------------+-----------+------------------+-------------+
|          1 | Aditya     | NULL        | Verma     | Sunrise Hostel   |         101 |
|          2 | Neha       | R           | Kulkarni  | Sunrise Hostel   |         101 |
|          4 | Sanya      | K           | Malhotra  | Blue Ridge Hostel |         101 |
|          5 | Vikram     | NULL        | Nair      | Blue Ridge Hostel |         102 |
|          6 | Ishita     | P           | Bansal    | Blue Ridge Hostel |         102 |
|          7 | Arjun      | NULL        | Reddy     | Lake View Hostel  |         101 |
+------------+------------+-------------+-----------+------------------+-------------+
6 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part B - Q2. List every staff member along with the name of their supervisor (show NULL/blank for staff with no supervisor).

SQL query:

```
SELECT e.Staff_ID, e.Staff_Name, e.Designation, m.Staff_Name AS Supervisor_Name
FROM staff e
LEFT JOIN staff m ON m.Staff_ID = e.Supervisor_ID
ORDER BY e.Staff_ID;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT e.Staff_ID, e.Staff_Name, e.Designat ion, m.Staff_Name AS Supervisor_Name
    -> FROM staff e
    -> LEFT JOIN staff m ON m.Staff_ID = e.Supervisor_ID
    -> ORDER BY e.Staff_ID;
+----------+---------------+-------------+-----------------+
| Staff_ID | Staff_Name    | Designation | Supervisor_Name |
+----------+---------------+-------------+-----------------+
|        1 | Ramesh Iyer   | Warden      | NULL            |
|        2 | Sunil Kumar   | Security    | Ramesh Iyer     |
|        3 | Geeta Rao     | Cleaner     | Ramesh Iyer     |
|        4 | Anita Sharma  | Warden      | NULL            |
|        5 | Mohan Das     | Security    | Anita Sharma    |
|        6 | Priya Menon   | Maintenance | Anita Sharma    |
|        7 | Kiran Bose    | Warden      | NULL            |
|        8 | Farhan Sheikh | Cleaner     | Kiran Bose      |
+----------+---------------+-------------+-----------------+
8 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part B - Q3. List every payment together with the paying student's name and the corresponding receipt number, if one has been generated.

SQL query:

```
SELECT p.Payment_ID, CONCAT(s.First_Name, ' ', s.Last_Name) AS Student_Name,
       p.Payment_Date, p.Amount, p.Payment_Type, r.Receipt_ID
FROM payment p
INNER JOIN student s ON s.Student_ID = p.Student_ID
LEFT JOIN receipt r ON r.Payment_ID = p.Payment_ID
ORDER BY p.Payment_ID;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT p.Payment_ID, CONCAT(s.First_Name, '  ', s.Last_Name) AS Student_Name,
    ->        p.Payment_Date, p.Amount, p.Payment_Type, r.Receipt_ID
    -> FROM payment p
    -> INNER JOIN student s ON s.Student_ID = p.Student_ID
    -> LEFT JOIN receipt r ON r.Payment_ID = p.Payment_ID
    -> ORDER BY p.Payment_ID;
+------------+----------------+--------------+----------+--------------+------------+
| Payment_ID | Student_Name   | Payment_Date | Amount   | Payment_Type | Receipt_ID |
+------------+----------------+--------------+----------+--------------+------------+
|          1 | Aditya Verma   | 2026-01-10   | 45000.00 | Online       |          1 |
|          2 | Neha Kulkarni  | 2026-01-11   | 45000.00 | Cash         |          2 |
|          3 | Rohan Shetty   | 2026-01-12   | 42000.00 | Card         |          3 |
|          4 | Sanya Malhotra | 2026-01-10   | 48000.00 | Online       |          4 |
|          5 | Vikram Nair    | 2026-01-13   | 48000.00 | Cheque       |          5 |
|          6 | Ishita Bansal  | 2026-01-14   | 48000.00 | Online       |          6 |
|          7 | Arjun Reddy    | 2026-01-15   | 40000.00 | Cash         |          7 |
|          9 | Kabir Oberoi   | 2026-08-22   | 45000.00 | Online       |          8 |
+------------+----------------+--------------+----------+--------------+------------+
8 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part B - Q4. Find every student who has never used any hostel facility.

SQL query:

```
SELECT s.Student_ID, s.First_Name, s.Last_Name
FROM student s
LEFT JOIN uses u ON u.Student_ID = s.Student_ID
WHERE u.Student_ID IS NULL;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT s.Student_ID, s.First_Name, s.Last_N ame
    -> FROM student s
    -> LEFT JOIN uses u ON u.Student_ID = s.Student_ID
    -> WHERE u.Student_ID IS NULL;
+------------+------------+-----------+
| Student_ID | First_Name | Last_Name |
+------------+------------+-----------+
|          8 | Meera      | Pillai    |
+------------+------------+-----------+
1 row in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part B - Q5. For each hostel, count how many students currently reside in it (include hostels with zero current residents).

SQL query:

```
SELECT h.Hostel_ID, h.Hostel_Name, COUNT(a.Student_ID) AS Current_Residents
FROM hostel h
LEFT JOIN allocation a ON a.Hostel_ID = h.Hostel_ID AND a.Check_out_Date IS NULL
GROUP BY h.Hostel_ID, h.Hostel_Name
ORDER BY h.Hostel_ID;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT h.Hostel_ID, h.Hostel_Name, COUNT(a. Student_ID) AS Current_Residents
    -> FROM hostel h
    -> LEFT JOIN allocation a ON a.Hostel_ID = h.Hostel_ID AND a.Check_out_Date  IS NULL
    -> GROUP BY h.Hostel_ID, h.Hostel_Name
    -> ORDER BY h.Hostel_ID;
+-----------+------------------+------------------+
| Hostel_ID | Hostel_Name      | Current_Residents |
+-----------+------------------+------------------+
|         1 | Sunrise Hostel   |                2 |
|         2 | Blue Ridge Hostel |                3 |
|         3 | Lake View Hostel  |                1 |
+-----------+------------------+------------------+
3 rows in set (0.002 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part B - Q6. List every facility usage record as: facility name, student name, usage date -- ordered by facility, then date.

SQL query:

```
SELECT f.Facility_Name, CONCAT(s.First_Name, ' ', s.Last_Name) AS Student_Name, u.Usage_Date
FROM uses u
INNER JOIN facility f ON f.Facility_ID = u.Facility_ID
INNER JOIN student s ON s.Student_ID = u.Student_ID
ORDER BY f.Facility_Name, u.Usage_Date;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT f.Facility_Name, CONCAT(s.First_Name , ' ', s.Last_Name) AS Student_Name, u.Usage_Date
    -> FROM uses u
    -> INNER JOIN facility f ON f.Facility_ID = u.Facility_ID
    -> INNER JOIN student s ON s.Student_ID = u.Student_ID
    -> ORDER BY f.Facility_Name, u.Usage_Date;
+---------------+---------------+------------+
| Facility_Name | Student_Name   | Usage_Date |
+---------------+---------------+------------+
| Common Room   | Arjun Reddy    | 2026-01-23 |
| Common Room   | Kabir Oberoi   | 2026-08-23 |
| Gym           | Sanya Malhotra | 2026-01-20 |
| Gym           | Vikram Nair    | 2026-01-21 |
| Gym           | Vikram Nair    | 2026-08-22 |
| Laundry Room  | Aditya Verma   | 2026-01-21 |
| Study Room    | Ishita Bansal  | 2026-01-22 |
| Study Room    | Sanya Malhotra | 2026-01-25 |
| WiFi Lounge   | Neha Kulkarni  | 2026-01-20 |
| WiFi Lounge   | Aditya Verma   | 2026-01-20 |
| WiFi Lounge   | Rohan Shetty   | 2026-01-22 |
| WiFi Lounge   | Kabir Oberoi   | 2026-08-22 |
+---------------+---------------+------------+
12 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part B - Q7. For every room, report how many beds in it are currently vacant.

SQL query:

```
SELECT r.Hostel_ID, r.Room_Number, COUNT(b.Bed_Number) AS Vacant_Beds
FROM room r
LEFT JOIN bed b ON b.Hostel_ID = r.Hostel_ID AND b.Room_Number = r.Room_Number
               AND b.Status = 'Vacant'
GROUP BY r.Hostel_ID, r.Room_Number
ORDER BY r.Hostel_ID, r.Room_Number;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT r.Hostel_ID, r.Room_Number, COUNT(b. Bed_Number) AS Vacant_Beds
    -> FROM room r
    -> LEFT JOIN bed b ON b.Hostel_ID = r.Hostel_ID AND b.Room_Number = r.Room_N umber
    ->                AND b.Status = 'Vacant'
    -> GROUP BY r.Hostel_ID, r.Room_Number
    -> ORDER BY r.Hostel_ID, r.Room_Number;
+-----------+-------------+-------------+
| Hostel_ID | Room_Number | Vacant_Beds |
+-----------+-------------+-------------+
|         1 |         101 |           0 |
|         1 |         102 |           2 |
|         1 |         201 |           1 |
|         2 |         101 |           1 |
|         2 |         102 |           0 |
|         3 |         101 |           1 |
+-----------+-------------+-------------+
6 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part B - Q8. List every staff member with the name and location of the hostel they work at.

SQL query:

```
SELECT st.Staff_ID, st.Staff_Name, st.Designation, h.Hostel_Name, h.Location
FROM staff st
INNER JOIN hostel h ON h.Hostel_ID = st.Hostel_ID
ORDER BY st.Staff_ID;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT st.Staff_ID, st.Staff_Name, st.Desig nation, h.Hostel_Name, h.Location
    -> FROM staff st
    -> INNER JOIN hostel h ON h.Hostel_ID = st.Hostel_ID
    -> ORDER BY st.Staff_ID;
+----------+--------------+-------------+------------------+--------------+
| Staff_ID | Staff_Name   | Designation | Hostel_Name      | Location     |
+----------+--------------+-------------+------------------+--------------+
|        1 | Ramesh Iyer  | Warden      | Sunrise Hostel   | North Campus |
|        2 | Sunil Kumar  | Security    | Sunrise Hostel   | North Campus |
|        3 | Geeta Rao    | Cleaner     | Sunrise Hostel   | North Campus |
|        4 | Anita Sharma | Warden      | Blue Ridge Hostel| South Campus |
|        5 | Mohan Das    | Security    | Blue Ridge Hostel| South Campus |
|        6 | Priya Menon  | Maintenance | Blue Ridge Hostel| South Campus |
|        7 | Kiran Bose   | Warden      | Lake View Hostel | East Campus  |
|        8 | Farhan Sheikh| Cleaner     | Lake View Hostel | East Campus  |
+----------+--------------+-------------+------------------+--------------+
8 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part B - Q9. For each hostel, compute the total fee revenue collected from its currently-residing students.

SQL query:

```
SELECT h.Hostel_ID, h.Hostel_Name, COALESCE(SUM(p.Amount), 0) AS Total_Revenue
FROM hostel h
LEFT JOIN allocation a ON a.Hostel_ID = h.Hostel_ID AND a.Check_out_Date IS NULL
LEFT JOIN payment p ON p.Student_ID = a.Student_ID
GROUP BY h.Hostel_ID, h.Hostel_Name
ORDER BY h.Hostel_ID;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT h.Hostel_ID, h.Hostel_Name, COALESCE (SUM(p.Amount), 0) AS Total_Revenue
    -> FROM hostel h
    -> LEFT JOIN allocation a ON a.Hostel_ID = h.Hostel_ID AND a.Check_out_Date  IS NULL
    -> LEFT JOIN payment p ON p.Student_ID = a.Student_ID
    -> GROUP BY h.Hostel_ID, h.Hostel_Name
    -> ORDER BY h.Hostel_ID;
+-----------+------------------+---------------+
| Hostel_ID | Hostel_Name      | Total_Revenue |
+-----------+------------------+---------------+
|         1 | Sunrise Hostel   |      90000.00 |
|         2 | Blue Ridge Hostel |     144000.00 |
|         3 | Lake View Hostel  |      40000.00 |
+-----------+------------------+---------------+
3 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part B - Q10. List every facility together with how many times it has been used, including facilities that have never been used.

SQL query:

```
SELECT f.Facility_ID, f.Facility_Name, COUNT(u.Student_ID) AS Times_Used
FROM facility f
LEFT JOIN uses u ON u.Facility_ID = f.Facility_ID
GROUP BY f.Facility_ID, f.Facility_Name
ORDER BY f.Facility_ID;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT f.Facility_ID, f.Facility_Name, COUN T(u.Student_ID) AS Times_Used
    -> FROM facility f
    -> LEFT JOIN uses u ON u.Facility_ID = f.Facility_ID
    -> GROUP BY f.Facility_ID, f.Facility_Name
    -> ORDER BY f.Facility_ID;
+-------------+---------------+------------+
| Facility_ID | Facility_Name | Times_Used |
+-------------+---------------+------------+
|           1 | WiFi Lounge   |          4 |
|           2 | Laundry Room  |          1 |
|           3 | Gym           |          3 |
|           4 | Study Room    |          2 |
|           5 | Common Room   |          2 |
+-------------+---------------+------------+
5 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part B - Q11. Write a query using a RIGHT JOIN between HOSTEL and FACILITY to list all facilities along with the name of the hostel providing each facility.

SQL query:

```
SELECT f.Facility_ID, f.Facility_Name, f.Operating_Hours, h.Hostel_Name
FROM hostel h
RIGHT JOIN facility f ON f.Hostel_ID = h.Hostel_ID
ORDER BY f.Facility_ID;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT f.Facility_ID, f.Facility_Name, f.Op erating_Hours, h.Hostel_Name
    -> FROM hostel h
    -> RIGHT JOIN facility f ON f.Hostel_ID = h.Hostel_ID
    -> ORDER BY f.Facility_ID;
+-------------+---------------+-----------------+-------------------+
| Facility_ID | Facility_Name | Operating_Hours | Hostel_Name       |
+-------------+---------------+-----------------+-------------------+
|           1 | WiFi Lounge   | 24 Hours        | Sunrise Hostel    |
|           2 | Laundry Room  | 06:00-20:00     | Sunrise Hostel    |
|           3 | Gym           | 05:00-22:00     | Blue Ridge Hostel |
|           4 | Study Room    | 24 Hours        | Blue Ridge Hostel |
|           5 | Common Room   | 08:00-23:00     | Lake View Hostel  |
+-------------+---------------+-----------------+-------------------+
5 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```

Part B - Q12. Write a query using a LEFT JOIN to list all students (First_Name, Last_Name, Email_Address) along with their assigned Room_Number. Ensure that students who do not have an allocated room, such as Rohan, are also included in the result.

SQL query:

```
SELECT s.First_Name, s.Last_Name, s.Email AS Email_Address, a.Room_Number
FROM student s
LEFT JOIN allocation a ON a.Student_ID = s.Student_ID AND a.Check_out_Date IS NULL
ORDER BY s.Student_ID;
```

Output:

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root) [smart_hostel]> SELECT s.First_Name, s.Last_Name, s.Email A S Email_Address, a.Room_Number
    -> FROM student s
    -> LEFT JOIN allocation a ON a.Student_ID = s.Student_ID AND a.Check_out_Dat e IS NULL
    -> ORDER BY s.Student_ID;
+------------+-----------+-----------------------+-------------+
| First_Name | Last_Name | Email_Address         | Room_Number |
+------------+-----------+-----------------------+-------------+
| Aditya     | Verma     | aditya.verma@pes.edu  |         101 |
| Neha       | Kulkarni  | neha.kulkarni@pes.edu |         101 |
| Rohan      | Shetty    | rohan.shetty@pes.edu  |        NULL |
| Sanya      | Malhotra  | sanya.malhotra@pes.edu|         101 |
| Vikram     | Nair      | vikram.nair@pes.edu   |         102 |
| Ishita     | Bansal    | ishita.bansal@pes.edu |         102 |
| Arjun      | Reddy     | arjun.reddy@pes.edu   |         101 |
| Meera      | Pillai    | meera.pillai@pes.edu  |        NULL |
| Kabir      | Oberoi    | kabir.oberoi@pes.edu  |        NULL |
+------------+-----------+-----------------------+-------------+
9 rows in set (0.001 sec)

(PES1UG24AM387@root) [smart_hostel]>
```